

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Opamp Gain Offset Design Utility Software

Enter component value (refer opamp datasheet)

Gain =	14
Offset (vols) =	0.5
The voltage level of a stable reference, Vref (vols) =	1
The Rf-scale output voltage, Voutf (vols) =	1
The non-zero output voltage, Voutzs (vols) =	0.3
The Rf-scale input voltage, Vinf (vols) =	0.1
The zero-scale input voltage, Vinzs (vols) =	0.05
Choose Rf Resistance, Rf (ohms) (Please insert a value) =	
Resistance R1 (ohms) =	1
Resistance R2 (ohms) =	1
Vref (vols) =	1
Resistance Rf (ohms) =	1

Fig. 3: Positive gain and negative offset

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Enter component value (refer opamp datasheet)

Gain =	-9
Offset (vols) =	3
The voltage level of a stable reference, Vref (vols) =	5
The Rf-scale output voltage, Voutf (vols) =	4
The non-zero output voltage, Voutzs (vols) =	3
The Rf-scale input voltage, Vinf (vols) =	0.05
The zero-scale input voltage, Vinzs (vols) =	0.1
Choose Rf Resistance, Rf (ohms) (Please insert a value) =	
Resistance R1 (ohms) =	9
Choose R2 Resistance, R2 (ohms) (Please insert a value) =	
Resistance Rf (ohms) =	9

Fig. 4: Negative gain and positive offset

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Enter component value (refer opamp datasheet)

Gain =	-28
Offset (vols) =	0.4
The voltage level of a stable reference, Vref (vols) =	5
The Rf-scale output voltage, Voutf (vols) =	3
The non-zero output voltage, Voutzs (vols) =	0.2
The Rf-scale input voltage, Vinf (vols) =	0.1
The zero-scale input voltage, Vinzs (vols) =	0.1
Choose Rf Resistance, Rf (ohms) (Please insert a value) =	
Resistance R1 (ohms) =	8
Resistance R2 (ohms) =	9

Fig. 5: Negative gain and negative offset

Fill all fields in the form in order to proceed to the next page, which is written in PHP, that is, opampCircuit.php.

The next page will show a recommended circuit design. This page also takes some inputs from you and calculates the required gain and offset values for circuit design by using the following relationship:

$$\text{Gain} = (\text{Voutfs} - \text{Voutzs}) / (\text{Vinf} - \text{Vinzs})$$

$$\text{Offset} = \text{Voutzs} - (\text{Gain} \times \text{Vinzs})$$

There will be four conditions, which are:

1. Positive gain and positive offset
2. Positive gain and negative offset
3. Negative gain and positive offset
4. Negative gain and negative offset

For positive gain and positive offset, output result of a typical example is shown in Fig. 2. Note the gain and offset values. Input parameters for this example are $V_{\text{ref}} = 5$, $V_{\text{outfs}} = 4$, $V_{\text{outzs}} = 3$, $V_{\text{inf}} = 2$ and $V_{\text{inzs}} = 1$.

Since the gain and offset values are both positive, the condition satisfies. Consult the datasheet to choose suitable resistor values for R_1 and R_f . As you enter these values, resistor values of R_2 and R_g will change automatically. Find the nearest resistances to use in the actual circuit and check the results.

For positive gain and negative offset, a sample output is shown in Fig. 3. Note the gain and offset values. Input parameters for this example are $V_{\text{ref}} = 5$, $V_{\text{outfs}} = 2$, $V_{\text{outzs}} = -0.3$, $V_{\text{inf}} = 0.1$ and $V_{\text{inzs}} = 0.05$.

Here, you need to provide only suitable R_f value. Remaining values are automatically calculated for the circuit design.

For negative gain and positive